C PROGRAMMING – ASSESSMENT 3

TASK 1 – Understand the Problem and Identify the Data (4 Marks)

Scenario: Grama Jyothi Rural Electricity Co-operative bills households based on meter readings and a fixed charge plus slab-wise energy charges.

1. Inputs and Outputs

Inputs: Previous meter reading, current meter reading, household number and consumer category. These identify the customer and help calculate usage and the applicable tariff.

Outputs: Units consumed, energy charge, fixed charge and total bill. The total bill is the sum of the fixed charge and energy charge.

2. Values That Remain Constant

The slab limits, tariff rates, fixed charge and meter capacity remain constant during one program run. They should be defined as named constants so that tariff changes can be made in one place.

3. Negative, Fractional and Large Values

Meter readings and units consumed should be non-negative. Money can contain fractions of a rupee, so calculations should use integer paise to avoid floating-point rounding errors. Household counts and accumulated billing values may be large, so an appropriate integer type such as long long should be considered where needed.

4. Rules for a Correct Bill

- The meter readings must be valid and within the meter's range.
- Meter rollover must be handled correctly when the reading changes from 65535 to 0.
- The first 100 units are free.
- Units 101–200 cost ■2.25 per unit.
- Every unit above 200 costs ■4.50 per unit.
- A fixed charge of ■50 is added.
- The total bill must not be negative and must equal the fixed charge plus the energy charge.

5. Assumptions

The meter is treated as a 16-bit unsigned counter with readings from 0 to 65535. After 65535, the next reading becomes 0. For one rollover, consumption can be calculated using unsigned 16-bit arithmetic. The tariff rates are assumed to remain unchanged during a single bill calculation.

Alternative Concept and Complexity Analysis

If meter readings were stored as text strings (for example, "01530"), each reading would first need to be converted into an integer before subtraction. For d digits, conversion takes O(d) time. A text representation uses space proportional to the number of digits, O(d), whereas an integer uses a fixed amount of space for the chosen type. Integer storage therefore makes repeated arithmetic simpler and more efficient.

Conclusion: Choosing suitable integer data types, storing money in paise, using named constants and handling meter rollover correctly helps produce accurate and reliable electricity bills.